CHE 321 - Exam 1 - Sample 1

exam1___1.m

```
function h = exam1(w,y)
X= y(1);
p= y(2);

% parameter
%since Ct0 given and only A then it must be Ca0
Ca0= 2.0;
Fa0= 10.0 ;
k=10.0 ;
alpha=.07;
eps= 3.0 ;


Ca=((Ca0*(1-X))*(1+eps*X))*p;
Cb=(Ca0*(1+X)/(1+eps*X))*p;
Cc=((Ca0*(3+3*X))/(1+eps*X))*p;


% rate eq
ra= -k*Ca-k*Cb*(Cc^3);

%eqns
h1= -ra/Fa0 ;
h2= (-alpha/2*p)*(1+eps*X);

h= [h1, h2]';

end
```

exam1_sim___1.m

```
clc;
clear all

% weight of particle we care about
wspan = [ 0 , 10];
% initial conditions for w and y
y0= [ 0 1];
% runs differential solver Rungekutta
[w,y] = ode45(@exam1,wspan,y0) ;

% Plot the results
figure(1);
plot(w, y(:, 1));

xlabel('Weight');
ylabel('conversion');
figure(2);
plot(w, y(:, 2));
xlabel('Weight');
ylabel('p');
title('Solution of ODE');

%trying to relate conversion to Dp
```

exam1_Q2___1.m

```
%plot conversion versus catalyste diameter ranging from .1mm to 10mm
%particle diameter means more void space
%more void space means more flow but also less surface area
% anytime you increase delta p with cause conversion loss
%having more void space means delta p decreases
%if delta p is decreasing, then conversion must be increasing

%given; Alpha is inversely proportional to Dp^2

%alpha= .07;
%p=1;
%esp=3;
% Define conversion range from 0 to 1
%conversion = h2; % Calculate conversion based on h2
%h2= (-alpha/2*p)*(1+eps*X);
%Dp= [.1, 10];

function h = exam1_Q2(w,y)
X= y(1);
p= y(2);

% parameter
%since Ct0 given and only A then it must be Ca0
Ca0= 2.0;
Fa0= 10.0 ;
k=10.0 ;
Dp= [.1, 10]
alpha= 1/Dp^2;
eps= 3.0 ;


Ca=((Ca0*(1-X))*(1+eps*X))*p;
Cb=(Ca0*(1+X)/(1+eps*X))*p;
Cc=((Ca0*(3+3*X))/(1+eps*X))*p;


% rate eq
ra= -k*Ca-k*Cb*(Cc^3);

%eqns
h1= -ra/Fa0 ;
h2= (-alpha/2*p)*(1+eps*X);

h= [h1, h2]';

end
```

Exam1_Q2_sim___1.m

```
clc;
clear all

% weight of particle we care about
wspan = [ 0 , 10];
% initial conditions for w and y
y0= [ 0 1];
% runs differential solver Rungekutta
[w,y] = ode45(@exam1_Q2,wspan,y0) ;

% Plot the results
figure(1);
plot(w, y(:, 1));
xlabel('Weight');
ylabel('conversion');
```